Physics-Grounded Generative HVAC Diagnosis

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Abstract

HVAC systems account for a significant portion of building energy use, yet their analysis remains largely expert-driven despite the availability of BAS data. This work develops a physics-grounded framework to analyze AHU operation by computing cooling load and fan energy from BAS data and interpreting these indicators using a large language model (LLM). Results from one year of data at NCSU's Fitts-Woolard Hall show that the system captures realistic operational patterns and generates consistent diagnostic insights, demonstrating the potential of combining physics-based analysis with generative AI for automated HVAC diagnostics.
GitHub: `https://github.com/ykim68ncstate/ECE592_GenAI-Introandapplication_Project_ykim68.git`

1 Introduction

Buildings represent a major portion of global energy consumption, and HVAC systems are among the most significant contributors to this demand. Despite the widespread availability of high-resolution Building Automation System (BAS) data, the analysis of HVAC operations remains largely dependent on expert knowledge. This limits scalability and prevents broader adoption of data-driven operational strategies.

This project aims to address this gap by developing a system that combines physics-based analysis with generative artificial intelligence. The core idea is to extract meaningful operational indicators from BAS data using physically grounded models and then leverage a large language model (LLM) to interpret these indicators in a human-readable form. By doing so, the system enables automated analysis of HVAC operations while maintaining physical interpretability.

2 Model Design

The overall framework consists of two tightly connected components: a physics-based analysis layer and a generative AI diagnostic layer

2.1 Physics-based Analysis Layer

The first component processes BAS data and computes key operational indicators. Sensor data from multiple souces are merged based on timestamps and aligned at a 15-minute resolution. When multiple sensors measure the same variable, such as return air temperature, their values are averaged to improve robustness. Cooling load is calculated using a standard air-side sensible heat equation:

$$Q = 1.08 \times CFM \times (T_{MA} - T_{SA})$$

This allows the system to quantify the thermal energy removed by the AHU. The load is converted to kW and integrated over time to obtain energy demand. Fan power is estimated using fan affinity laws, where power scales with the cube of the fan speed. Each fan's rated brake horsepower is converted to electric input power, and total fan power is obtained by summing across all fans. This provides a physically consistent estimate of fan energy consumption.

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2.2 Generative AI Diagnostic Layer

The generative AI diagnostic layer is built using the OpenAI GPT-5 model, accessed via the OpenAI API. Rather than directly processing raw time-series data, the model takes structured summaries of AHU operation as input. These summaries include physically interpretable variables such as cooling load, airflow, temperatures, and estimated fan power. Based on this input, the model generates human-readable diagnostic reports that describe system operation, interpret cooling demand, evaluate fan energy usage, and provide simple recommendations.

This design allows numerical outputs to be translated into actionable insights, enabling automated HVAC operation analysis.

3 Dataset and Processing

The dataset used in this study consists of one year of Building Automation System (BAS) data collected from the Fitts-Woolard Hall building at North Carolina State University (NCSU). The dataset covers a continuous period from January 1, 2025 to December 31, 2025, and includes high-resolution measurements at 15-minute intervals.

The data include multiple sensor streams related to AHU operation, such as outdoor air temperature, return air temperature, mixed air temperature, supply air temperature, outdoor airflow, supply airflow, and fan speed signals for multiple fans. Each variable was originally stored in separate files corresponding to individual measurement points.

To construct a unified dataset, all sensor data were merged using their timestamps and aligned to a consistent 15-minute time resolution. Data preprocessing included timestamp conversion, sorting, and synchronization across all variables. For variables with multiple sensors, such as return air temperature, measurements were averaged to obtain a representative value and improve robustness.

4 Experimental Results and Analysis

The implemented system successfully computes key operational indicators from BAS data and reveals physically meaningful patterns in AHU behavior over time.

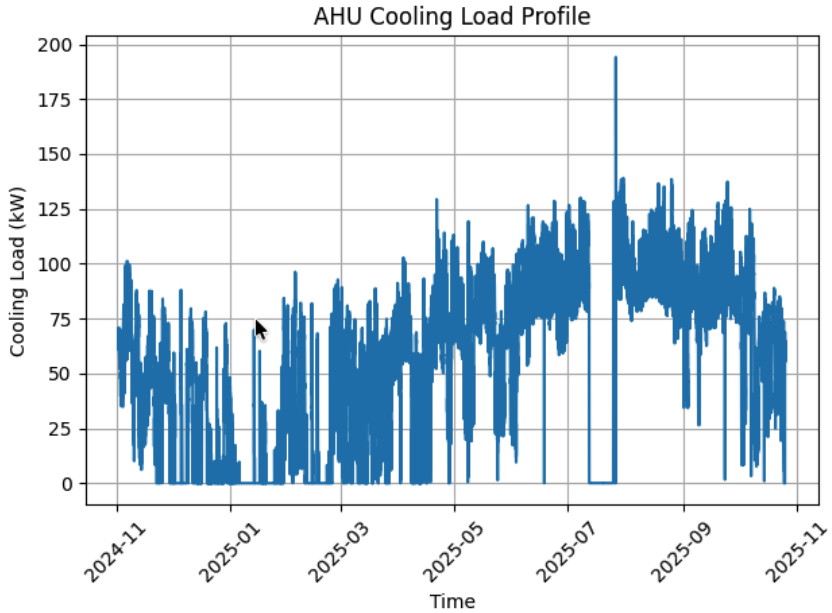

Figure 1: Time-series cooling load profile of the AHU showing seasonal variation and peak demand behavior.

Figure 1 presents the time-series cooling load profile of the AHU over the analyzed period. The cooling load varies significantly throughout the year, reflecting seasonal changes and varying operating conditions. Higher cooling demand is observed during warmer periods, where both the temperature difference between mixed air and supply air and the airflow rate increase. The peak cooling load approaches approximately 200 kW, while lower loads are observed during transitional seasons and nighttime operation. These trends are consistent with expected thermodynamic behavior, confirming that the cooling load calculation captures realistic system dynamics.

The relationship between cooling load and airflow can be further understood through the computed operational indicators. Periods of higher cooling load generally correspond to increased supply airflow and fan operation, indicating coordinated system control. The estimated fan power, derived using fan affinity laws, reflects this behavior, showing increased energy consumption during high-demand periods.

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1) Operational summary
- Cooling mode, SAT 54.9 F with 14.4 kcfm supply.
- OA flow 5.30 kcfm (~37% OA). OAT 65.9 F, RAT 71.5 F, MAT 69.9 F (mixing aligns with OA fraction).
- Coil ΔT (MAT→SAT) ≈ 15.0 F; sensible cooling ≈ 233.8 kBtu/h (~19.5 tons, 68.5 kW).
- Interval energy (15 min): cooling 17.1 kWh; fan 1.21 kWh.

2) Cooling demand interpretation
- Moderate mechanical cooling for late evening conditions; OAT < RAT but > SAT, so economizer alone can't meet 55 F SAT
- With current OA fraction, MAT is near return temp; moving toward 100% OA would drop MAT to ~65.9 F, cutting coil load

3) Fan power interpretation
- Estimated fan power 4.85 kW → ~0.34 kW/kcfm (0.34 W/CFM), which is efficient and consistent with the airflow; no fan

4) Operational status
- Appears normal: temperatures are self-consistent with airflow and load; energy aligns with a 15-minute interval; mixi

5) Recommended action
- Verify economizer enable/enthalpy high-limit and OA damper command. If OA enthalpy is below return and low-limit/humi
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Figure 2: Example of LLM-generated diagnostic report based on structured AHU operational summary.

Figure 2 illustrates an example of the LLM-generated diagnostic output based on a single time-step summary. The generated report provides a structured interpretation of AHU operation, including operational conditions, cooling demand assessment, fan power evaluation, and recommended actions. The output demonstrates that the LLM is capable of translating numerical indicators into concise and meaningful engineering insights.

	timestamp	mode	cooling_load_kw_thermal	total_fan_power_kw_est	report
0	2024-11-01 23:01:13	cooling	68.52	4.85	1) Operational summary\n- AHU in cooling at ni...
1	2024-11-01 23:16:13	cooling	64.91	4.57	1) Operational summary\n- Mode: Cooling at nig...
2	2024-11-01 23:31:13	cooling	61.79	3.89	1) Operational summary\n- Mode: Cooling; SAT 5...
3	2024-11-01 23:46:13	cooling	70.91	5.33	1) Operational summary\n- Mode: Cooling; SAT 5...
4	2024-11-02 00:01:13	cooling	63.09	4.19	1) Operational summary\n- AHU in cooling at ni...

Figure 3: Sample outputs showing multiple time-step summaries and corresponding LLM-generated interpretations.

Figure 3 presents a tabular view of multiple time-step summaries alongside the corresponding LLM-generated reports. The results show that the model produces consistent and coherent interpretations across different operating conditions. The generated outputs align with the underlying physical relationships observed in the data, such as the dependency of cooling load on temperature difference and airflow.

5 Current Progress

In this milestone stage, the core pipeline of the proposed system has been constructed. The system processes BAS data, computes physics-based operational indicators such as cooling load and fan energy, and generates interpretable summaries using a large language model (LLM). These results show that AHU operational data can be transformed into meaningful insights through a combination of physical modeling and generative AI.

6 Next Steps

For the final version, several enhancements are planned. First, the system will be extended to evaluate AHU operation based on HVAC control standards, particularly ASHRAE Guideline 36, enabling the LLM to assess whether current operation is appropriate rather than only describing it. Second, the LLM component will be further refined to provide more structured and reliable diagnostic reasoning, including clearer separation between description, evaluation, and recommendation. Third, the system will support interactive, question-based communication, allowing users to query AHU operation in a conversational manner. Finally, the model will be extended to provide actionable guidance for energy savings, suggesting how system operation can be improved to reduce energy consumption.